Accelerated Semiconductor R&D: AI Surrogate Modeling

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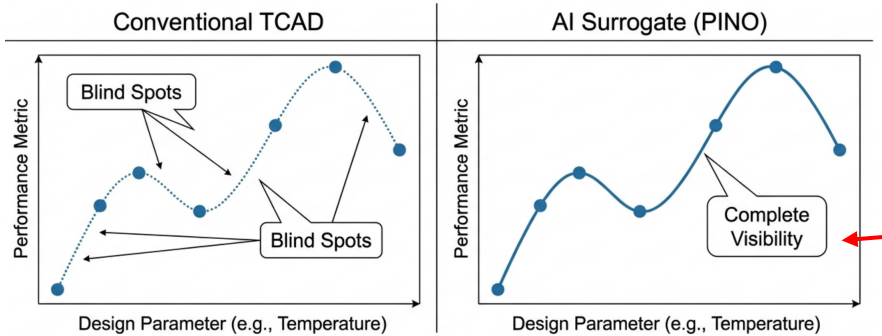
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Physics-Informed AI to break through R&D Bottleneck

- **The Cost of Precision:** Conventional TCAD simulations for advanced Memory (NAND/DRAM) are computationally prohibitive (hours/days per run).
  - **Optimization "Blind Spots":** Sparse data points lead to missed opportunities in design space exploration between simulated nodes.
  - **Static Methodology:** Altering a single geometry parameter (e.g., capacitor aspect ratio) requires a total mesh rebuild and re-simulation.
- **Beyond "Black Box" AI:** Implementing PINO (Physics-Informed Neural Operators)
  - Introducing Physics Loss (Poisson, Arrhenius, etc.) within the model architecture.



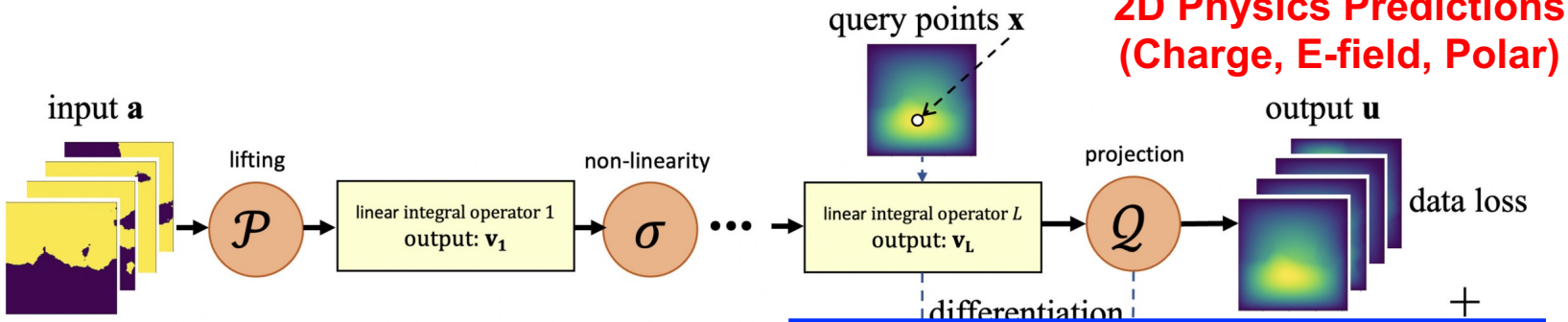
$$\mathcal{L}_{total} = \underbrace{\|\hat{u} - u_{target}\|_{L_1}}_{\text{Data Accuracy}} + \lambda \cdot \underbrace{\|\nabla\hat{\phi} - \nabla\phi_{target}\|_{L_1}}_{\text{Physical Consistency}}$$

(Matching sparse TCAD ground truth) (Penalizing PDE violations)

PINO Architecture

- Training Data: Fully compatible with conventional TCAD.

Output:
2D Physics Predictions
(Charge, E-field, Polar)



Input:
Device Geometry
Experimental Condition
Material Properties

Physics-informed
Loss function

e.g., Poisson Equation (Charge-Field Derivative Relationship)

Zongyi Li et al, Physics-Informed Neural Operator for Learning Partial Differential Equations

Simple Example: 3D NAND Charge Migration Modeling

- **Retention loss due to charge migration**
  - **Input Variables (according to IRPS 2019):**
    - **1. Tunnel layer Thickness (t_{TL})**
      - t_TL = 2, 2.5, 3, 3.5, 4nm to
    - **2. PGM voltage (V_{PGM})**
      - V_PGM = 10, 15, 20V
    - **3. Cell Spacing (L_{SP})**
      - L_sp = 20, 25, 30nm

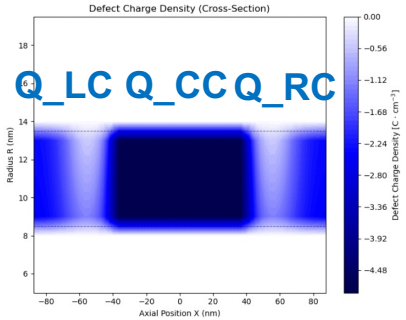
PINO Map →

$$Q = \int \rho_{def}(r,t)dr$$

Output: Retention Charge Map

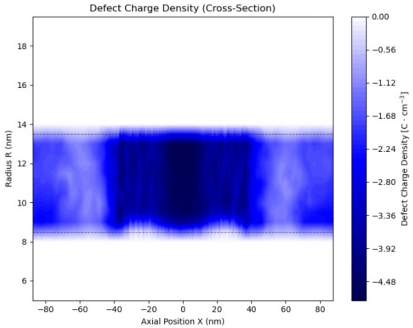
→ $Q_{Migrate} = \Delta(Q_{LC}+Q_{RC})[t_{ret} - t_0]$

$w.r.t.\{t_{TL}, V_{PGM}, L_{SP}\}$

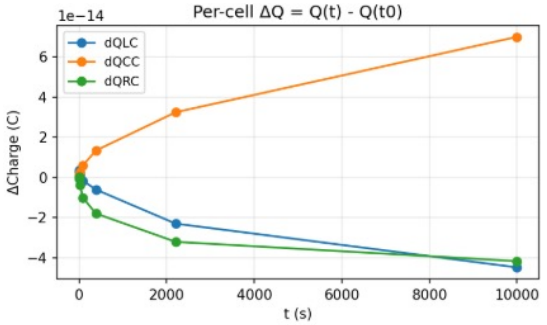


t=0

Retention Loss →



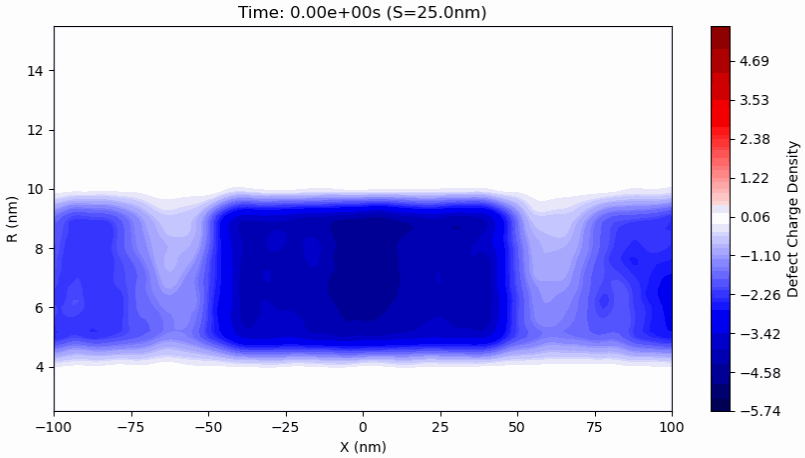
t=1e4s (=t_ret)

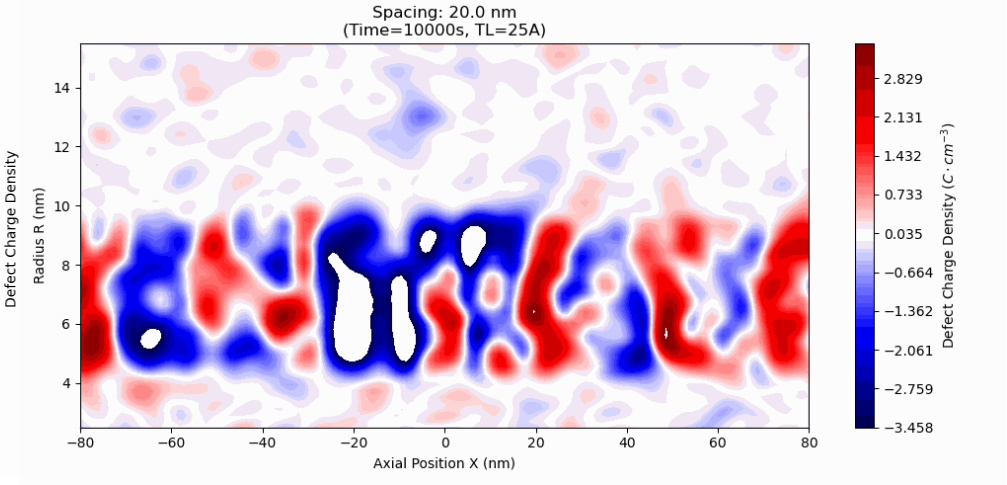


Calculated Retention Loss

3D NAND Charge Migration Modeling

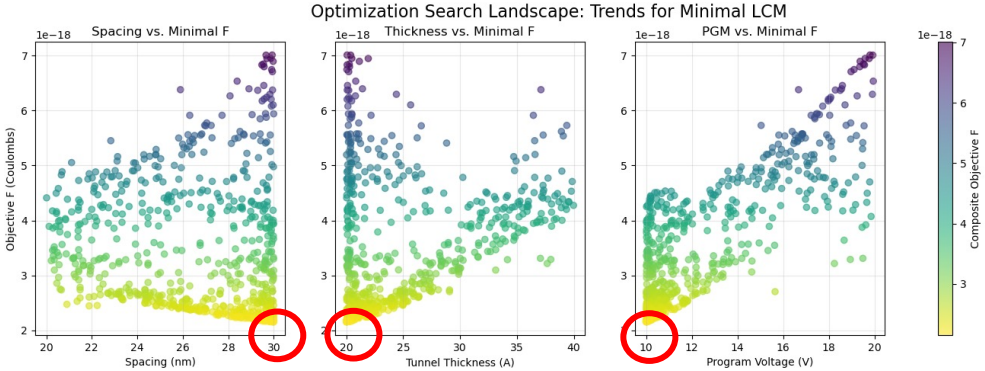
- **Advantage 1) Accelerated Simulation**
  - Interpolation visualization available
  - Time advantage: 3 days+ for discrete sweeps vs. inference less than milliseconds

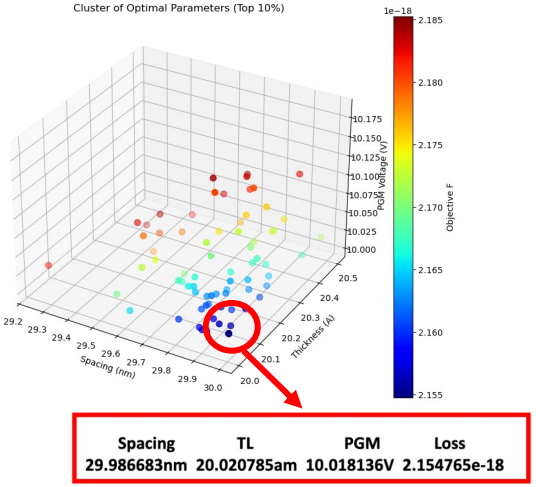




3D NAND Charge Migration Modeling

- **Advantage 2) Multi-parameter Optimization**
  - Can observe physically reliable result
  - Can expand this framework for non-linear variables optimization



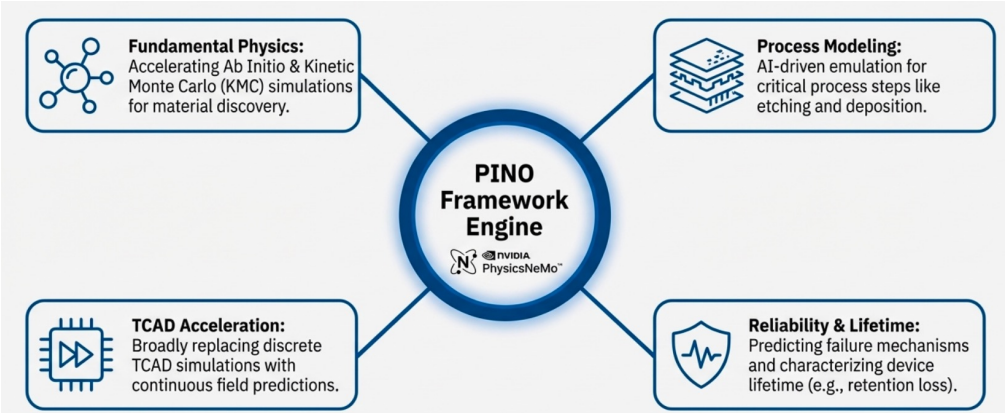


Variables with physical interpretation (IRPS 2019):

1. Wide Cell Spacing → Prevents LCM
2. Thick Tunnel Layer Thickness → LCM ratio increases
3. High PGM Voltage → LCM-dominant

Application on Memory R&D

- **Reliability Forecasting:** Accelerated modeling of time-dependent degradation.
- **Virtual Corner-Case Testing:** Rapidly identify failure points at extreme operating conditions without exhausting lab resources.
- **Design Optimization:** Fine-tune critical dimensions (pitch, channel diameter) through thousands of virtual iterations in minutes.
- **Accelerated Time-to-Market:** Bridges the gap between initial concept and silicon validation by streamlining the pre-silicon optimization loop.



Thank you!

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